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ORIGINAL RESEARCH



Description of the 2020 NEMSIS Public-Release Research Dataset

Julianne Ehlers , Benjamin Fisher , Skyler Peterson, Mengtao Dai, Angela Larkin, Lauri Bradt, and N. Clay Mann 

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ABSTRACT

Background: The National Emergency Medical Services Information System (NEMSIS) is a federally funded program designed to standardize Emergency Medical Services (EMS) patient care reporting and facilitate state and national data repositories for the assessment and improvement of EMS systems of care. This manuscript characterizes the 2020 submissions to the National EMS Database, detailing the strengths and limitations associated with use of these data for public health surveillance, improving prehospital patient care, critical resource allocation, clinician safety, system quality assurance and research purposes.

Methodology: Using the 2020 NEMSIS Public-Release Research Dataset (NEMSIS dataset), we evaluated the dataset completeness (i.e., presence of missing/null values), dataset content and assessed data generalizability. The analysis focused on 9-1-1 EMS activations resulting in the treatment and transport of a patient, except for out-of-hospital cardiac arrests for which all patients were included regardless of transport status.

Results: In 2020, 43,488,767 EMS activations were reported to the National EMS Database by 12,319 agencies serving 50 states and territories. Of the 19,533,036 9-1-1 EMS activations reportedly treating and transporting a patient, the majority were attended by “non-volunteer” clinicians (77%) working in a fire-based EMS agency (35%) certified to offer Advanced Life Support (ALS) Paramedic service (80%) and located in an urban area (82%). 9-1-1 call centers most often dispatched EMS for “sick person” (20%), while EMS clinicians most likely reported asthenia (7%) as the patient’s primary symptom as well as the clinician’s primary impression (6%), and documented “fall on same level, slip, or trip” as the most common cause of injury (37%). The NEMSIS dataset demonstrates some “missingness” and element inconsistencies, but methods may be employed to mitigate these data limitations.

Conclusions: The National EMS Database is a free and publicly available resource for evaluating EMS system utilization, response, and prehospital patient care. Understanding the characteristics of the underlying dataset and known data limitations will help ensure proper analysis and reporting of research and quality metrics based on nationally standardized NEMSIS data.

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Introduction

Since the introduction of structured emergency medical services in the U.S., there has been a call for the systemization of collected data to characterize the emergency care provided in the out-of-hospital setting (1). In 1973, the Emergency Medical Services Systems Act was passed, which in part, was enacted to initiate a standardized format and process for documenting and dispensing emergency care. This was the first legislation to require data or documentation of EMS services (2). The first major document to publish data points that specifically address a segment of EMS care was the Utstein Style for Uniform Reporting of Data, published in 1991 (3). These data points, specific to cardiac arrest care, also included elements assessing hospital patient outcome, which largely remain unavailable to EMS personnel (4). Efforts are currently underway to improve the availability of patient outcome data in a systematic manner (5).

Over the years, sporadic reports in the popular press suggested that data elements designed to assess similar attributes of EMS system response or care varied widely across EMS systems (6). In 1993, the National Highway Traffic Safety Administration (NHTSA) developed a national consensus document that defined 81 elements considered essential (or important) to capture in EMS information systems (7). This document was referred to as the NHTSA Uniform Prehospital EMS Dataset Version 1.0. Soon after, NHTSA published the 1996 *EMS Agenda for the Future. A Vision for the Nation’s EMS System*, which outlined 5 recommendations for EMS information systems (8). The National Center for Injury Prevention and Control followed suit and published the *Data Elements for Emergency Department Systems* which extended the concept of an information system to the

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Table 1. Summary of EMS data submission progression.

Statistical Year	Reporting States/Territories ¹	Reporting Agencies	Number of Activations	Treated and Transported 9-1-1 Response ²
2009	26	2,112	5,767,090	3,367,668
2010	31	4,874	9,874,748	4,874,061
2011	35	7,180	14,371,941	7,701,605
2012	43	8,439	19,831,189	10,733,925
2013	45	9,025	23,897,212	12,595,958
2014	48	9,653	25,835,729	13,769,286
2015	49	10,137	30,206,450	15,729,516
2016	49	9,993	29,919,652	15,361,777
2017	35	4,016	7,907,829	4,233,046
2018	43	9,599	22,532,890	10,675,178
2019	47	10,062	34,203,087	17,089,547
2020	50	12,319	43,488,767	19,533,036

¹Number of reporting states and territories of the United States. States and territories that did not contribute data to the 2020 Public-Release Research Dataset are Delaware, Idaho, Hawaii, Massachusetts, American Samoa, and Puerto Rico. Dataset reference material: <https://nemsis.org/using-ems-data/request-research-data/>.

²Only including the activations that are 9-1-1 calls, treated and transported by EMS.

emergency department and provided standards for data collection and purported linkages back to EMS (9).

Beginning in 2001, The National Association of State EMS Officials, in conjunction with its federal partners at NHTSA and the Trauma/EMS Systems program of the Health Resources and Services Administration (HRSA), began work to develop a national EMS database—known as NEMSIS (10). To initiate this work, state and territory representatives were asked to sign a “Memorandum of Understanding” which recognized the need for standardized EMS data collection at the state level as well as the assignment of a specific set of data elements to be collected on a national level. By 2008, all states and territories had signed the memorandum. Furthermore, during this time period, a fully vetted data dictionary (NHTSA Uniform Prehospital EMS Dataset Version 2), including an XML-based architecture supporting relationships between elements, was completed and made available to EMS software developers to ensure market saturation and consistency in application.

In 2007, a national EMS dataset was made accessible to the public, including 300,000 EMS activations submitted by 3 states. Early analysis of this NEMSIS version 2 dataset revealed that revisions could improve the integrity of the resulting data and work on version 3 of the NEMSIS data standard was initiated. During 2016, the NEMSIS version 3 data standard underwent adoption across the states. Through 2017, there was a decrease in data submissions as states voluntarily and without federal funding implemented the NEMSIS version 3 data standard which entailed updating their state resources, infrastructure, software upgrades, and extensive training (Table 1).

Purpose and Objectives

The purpose of this work is to describe characteristics of the NEMSIS dataset and highlight the availability of public EMS data to be utilized in the improvement of EMS response, public health surveillance, improving prehospital patient care, critical resource allocation, clinician safety, system quality assurance and research purposes.

The objectives were to provide information regarding the validity and reliability of data, discuss the potential for bias,

and report descriptive analyses of key variables from the 2020 calendar year for 9-1-1 initiated EMS activations that result in a treated and transported patient.

Methods

Study Design

We analyzed data contained in 2020 NEMSIS Public-Release Research Dataset (i.e., NEMSIS dataset) archived in the National EMS Database housed at the University of Utah School of Medicine Data Coordinating Center.

Data Source

The NEMSIS dataset represents a large convenience sample of EMS activations, maintained by the NEMSIS TAC. State/territory Offices of EMS electronically submit de-identified data from patient care reports (PCR) to the TAC which have been generated by local EMS agencies submitting to state/territory data repositories. The data released for public use (i.e., NEMSIS dataset) are nationally aggregated and geographic/agency-specific identifiers are removed prior to public distribution.

The data reflected in this report were primarily submitted in the NEMSIS v3.4 standard (11). Of the 585 data elements available in the v3.4 standard, 165 are national elements and collected by the NEMSIS TAC. Each state/territory determined which additional data elements they require collected, and EMS agencies may also implement the collection of elements specific to their service. State specific elements are not submitted to the National EMS Database but are archived at the state/territory level.

Much care is taken to ensure that the data are as clean as possible. All data submitted from states/territories must comply with the NEMSIS XML standard and are exposed to several hundred data validation rules checking for completeness, logical consistency, and proper formatting. Submissions that do not pass validation are rejected or flagged. For the most part, the NEMSIS TAC does not correct any remaining identified errors. The TAC does not require states or agencies to correct identified errors retrospectively, but errors are reported to states, to promote

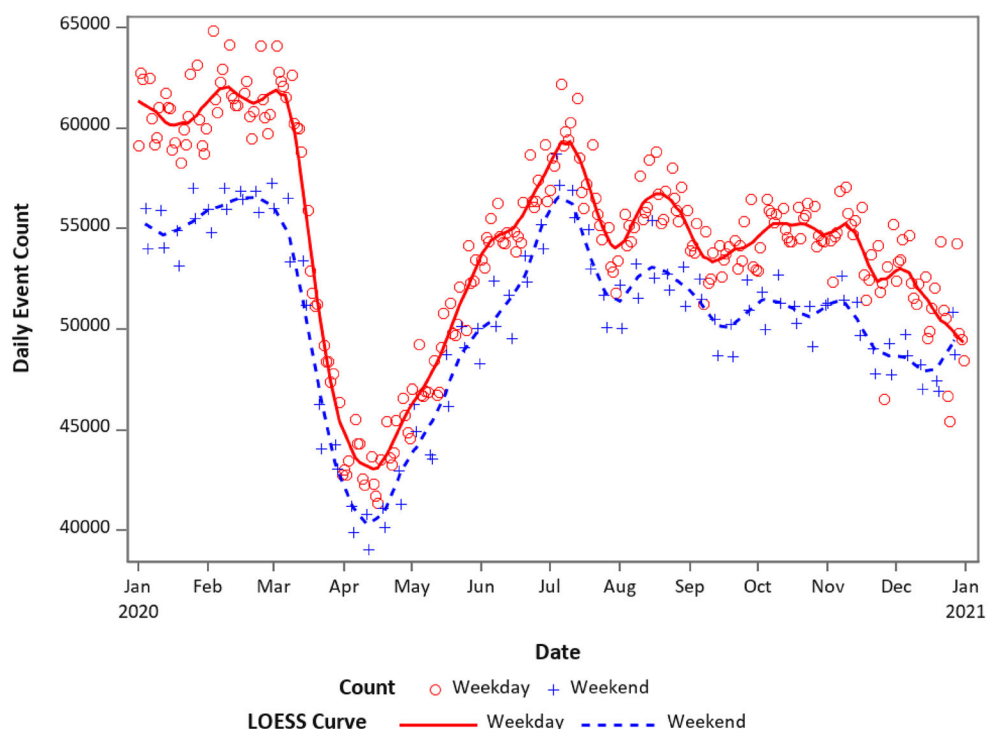


Figure 1. Daily Average Number of EMS Activations Resulting in Treatment and Transport.

*Lines represent a locally estimated scatterplot smoothing (LOESS) method.

future data refinement. Thus, the information contained in the data are provided as reported by states/territories with inherent deficiencies.

Selection of Cases

The inclusion criteria utilized by states/territories to characterize EMS activations aggregated in state dataset repositories vary. For example, states/territories may mandate every EMS activation be reported, while other states report only activations resulting in patient contact. This variation is reflected in the constitution of the NEMSIS dataset. It is also important to note that current national and state policies do not facilitate a process for tracking an individual patient through an emergent health care event. This results in the possibility of multiple PCRs for the same patient, reported by multiple EMS resources, submitted to the state repositories, and thus, the National EMS Database. Therefore, the NEMSIS dataset should be viewed as a registry of EMS activations (or responses), rather than a registry of patients receiving emergency care. This analysis focused on 9-1-1 EMS activations resulting in the treatment and transport of a patient, except for out-of-hospital cardiac arrests where all patients were included regardless of transport status.

Outcomes and Covariates

The NEMSIS dataset includes a collection of elements designed to characterize the health outcome of patients, based upon information collected during an emergency department or hospital stay. To date, however, these

elements are scarcely populated. Nevertheless, the care provided by EMS professionals in the out-of-hospital setting are well documented, allowing for the assessment of system responsiveness, medication/procedure use and effectiveness, protocol adherence, etc. In addition, patient characteristics and detailed attributes of the underlying illness or injury are documented allowing for statistical modeling, accounting for a variety of patient and emergency event attributes. Other data available for assessment include EMS agency characteristics, patient complaint reported to 9-1-1 call centers, delays experienced in the EMS response, identification of mass casualty events and patient disposition decisions.

Data Analysis

We analyzed EMS response elements and patient demographics using descriptive statistics, including common performance metrics, to characterize the dataset and assess the quality of the individual data points. To approximate the urbanicity of EMS agency service areas, agency home ZIP Codes were associated with United States Department of Agriculture (USDA) Urban Influence Codes (12). Database management and descriptive analyses were conducted using SAS (v 9.4, SAS Institute, Cary, NC).

Results

In 2020, a total of 12,319 EMS agencies, serving in 50 states and territories submitted 43,488,767 EMS activations to the National EMS Database (Table 1). Of these, 19,533,036 EMS activations were dispatched because of a 9-1-1 call and resulted in the treatment and transport of a patient to a health

Table 2. Distribution of 9-1-1 EMS Activations by Agency Level of Service.

Level of Service ¹	Overall N = 19,533,036
ALS-Paramedic	15,591,274 (80%)
BLS-Basic /EMT	2,881,779 (15%)
ALS-AEMT	388,870 (2%)
Specialty Critical Care	297,782 (2%)
ALS-Intermediate	258,221 (1%)
BLS-AEMT	66,536 (<1%)
ALS-Nurse	16,867 (<1%)
BLS-First Responder/EMR	14,532 (<1%)
BLS-Intermediate	8,667 (<1%)
ALS-Community Paramedicine	4,924 (<1%)
ALS-Physician	2,930 (<1%)
BLS-Community Paramedicine	654 (<1%)

¹Based on eResponse.15-Level of Care of This Unit.

Results reflect data as of 2020 public data, treat and transported 9-1-1 response. ALS (Advanced Life Support).

BLS (Basic Life Support).

EMT (Emergency Medical Technician).

AEMT (Advanced Emergency Medical Technician).

EMR (Emergency Medical Responder).

care facility. More 9-1-1 initiated EMS activations occur on weekdays, compared to weekends (Figure 1). In 2020, the expected distribution of activations was acutely disrupted by the COVID-19 pandemic compared to previous years (13).

Agency/System Characteristics

Of the 19,533,036 treat and transport 9-1-1 EMS activations occurring in 2020, the vast majority involved service provided by EMS agencies credentialed to provide ALS-Paramedic level service (80%) (Table 2). Based upon USDA Urban Influence Codes (12), 82% of reporting EMS agencies reside in urban areas, 6% in suburban, 7% in rural and 2% in frontier areas. When considering the organizational status of EMS agencies, 77% of EMS activations were initiated by non-volunteer organizations, 21% by mixed paid/unpaid organizations, and 2% by strictly volunteer EMS organizations. Fire department-based EMS (35%) responded most often, followed by private, non-hospital EMS organizations (31%), governmental organizations (25%), hospital-based (8%) and tribal nations EMS units (<1%) (Table 3).

Cumulative elapsed EMS response times demonstrate the expected Poisson distributions (Figure 2). The median elapsed time from “unit en route” to “arrival on-scene” was 7 min (IQR, 6 min). The cumulative time from “unit en route” to “depart scene” was 23 min (IQR, 13 min) and cumulative time to patient arrival at destination was 36 min (IQR, 20 min). Each individual distribution demonstrates the same area under the curve, but distributions sequentially become less “peaked”, most likely in response to the increasing unpredictability and variation of on-scene and transport environments.

Patient Condition Characteristics

Among 9-1-1 initiated EMS responses resulting in the treatment and transport of a patient, sick person (20%) was the most frequent patient complaint reported to EMS by dispatch, followed by breathing problem (11%), a fall victim (9%), and chest pain (7%) (Table 4). Once on the scene, the

Table 3. Distribution of 9-1-1 EMS Activations by Agency Status, Type, and Urbanicity Overall N = 19,533,036.

Organization Status (dAgency.12)	
Non- Volunteer	14,966,828 (77%)
Mixed	4,135,849 (21%)
Volunteer	430,359 (2%)
Organization Type (dAgency.13)	
Fire Department	6,924,622 (35%)
Private Non-Hospital	6,101,068 (31%)
Governmental	4,969,809 (25%)
Hospital	1,503,103 (8%)
Tribal	34,434 (<1%)
Urbanicity ¹	
Urban	16,088,286 (82%)
Rural	1,344,865 (7%)
Suburban	1,215,616 (6%)
New/Unregistered	564,627 (3%)
Frontier	319,642 (2%)

¹Urbanicity is applied after data collection.

Results reflect data as of 2020 public data, treat and transported 9-1-1 response.

primary patient symptom reported by EMS included asthenia (7%), shortness of breath (7%) and generalized pain (6%) (Table 5). The clinician’s primary impression of the patient’s condition most often cited was asthenia (6%), generalized abdominal pain (5%), chest pain (5%), and injury (5%) (Table 6).

Characteristics of Cardiac Arrest

During 2020, 448,684 EMS activations involved a cardiac arrest (Figure 3). Eighty-eight percent of activations reported an arrest occurring prior to EMS arrival. When these activations were stratified by offered prior aid (by law enforcement, lay person, or other clinician), interesting relationships appear in these data. For example, PCRs recording the application of prior aid demonstrate fewer “deaths on scene” and more hospital transports compared to activations reporting no prior aid. However, limitations in the NEMSIS dataset do not allow for other stratifications, such as elapsed time since arrest to further explain the observed relationships. Nevertheless, in this instance, preliminary associations can be assessed or even refined with the use of related NEMSIS elements and statistical modeling.

Characteristics of Traumatic Injuries

Based upon patient complaints reported to 9-1-1 call centers, injury (when combining traumatic and non-traumatic injury) was a major recorded reason for seeking emergency care. Once EMS arrived on scene, clinicians documented the cause of injuries, suggesting that the bulk of EMS activations involved patients suffering a fall (47%), motor vehicle collision (17%), or were involved in an assault (7%) (Table 7). EMS clinicians reported that for nearly 15% of injury-related activations, patients suffered major trauma, 12% experienced severe pain and 7% had a suspected fracture/dislocation (data not shown). Only 3 injury types represent more than 1% of all injuries, demonstrating the diversity of reported injuries. For 9% of activations reporting a possible injury (based on an affirmative response to eInjury.01), clinicians did not document a cause.

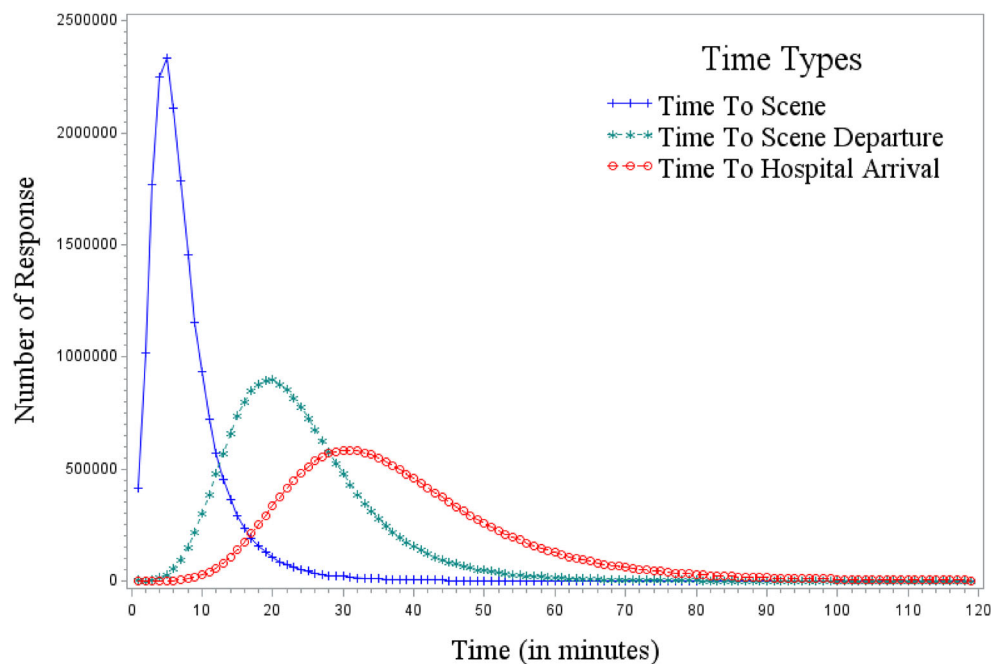


Figure 2. Frequency of Elapsed EMS System Time.

Characteristics of EMS Activations Associated with COVID-19

EMS activations resulting in the care of patients with influenza-like illness (ILI) closely correlated with confirmed COVID-19 cases documented in other sources (14). Due to the lack of specific COVID-19 related ICD-10-CM codes at the beginning of the pandemic, existing codes describing ILI were captured based on EMS clinician's primary or secondary impression, and/or documented primary or secondary symptoms (Figure 4).

The percentage of 9-1-1 initiated EMS activations resulting in contact with a patient presenting with ILI dramatically increased in week 10 of 2020 which was associated with the CDC announcement of community spread of COVID-19 (15). Two weeks after states began shutting down to prevent transmission (week 12), the percentage of ILI related EMS activations began to subside. Additional spikes in the percentage of ILI-related EMS activations quickly followed summer holiday weekends (i.e., Memorial Day and Labor Day) and steadily increased with the beginning of the traditional flu season. Additional disease and injury types that demonstrated notable trend inflections included substantial increases in EMS activations for cardiac arrest, potential drug overdose and mental/behavioral issues with sizable reductions in EMS activations for patients with injuries (data not shown) (14).

Data Quality

Data elements found on EMS patient care reports with finite code sets are most often "complete" when submitted to the National EMS Database (e.g., patient sex). NEMSIS elements relying on "lists" (e.g., cause of injury) are more likely incomplete or less descriptive (Figure 5). The accuracy of EMS clinician documentation was greatly enhanced, beginning in 2017, with the introduction of validity rules that

present to the clinician at the time of patient care documentation.

Discussion

The NEMSIS dataset includes approximately 19.5 million 9-1-1 activations involving the treatment and transport of a patient. Data from the 2018 National Hospital Ambulatory Medical Care Survey (NHAMCS) suggests that approximately 20.8 million patients arrived at hospitals via ambulances (16). Based upon these estimates, the NEMSIS dataset may be approaching a complete census, able to represent the 9-1-1 emergency response system in the U.S. The vast majority of federally funded health care datasets rely on methods to systematically sample patients to represent the underlying patient populations and are therefore inherently restricted in their ability to generalize to the full spectrum of services (and patient populations) accessing and receiving care in the United States.

Nevertheless, the fact that the NEMSIS dataset represents a "convenience sample" introduces the possibility that various forms of bias may be inherent to the sample. With submissions to the National EMS Database during the initial years, it was clear that states and agencies with an exceptional commitment to monitoring and improving system performance and patient care were reporting. As the commitment to the National EMS Database increases among and within states, the potential effect of this form of selection bias decreases. Thus, selection bias should be considered in analyses due to the potential for inconsistency in documentation among clinicians with varying degrees of medical training and differences in treatment and transport policies and protocols. The potential for information bias also exists as many NEMSIS elements allow the use of null values. These data limitations may systematically limit the ability to compare system response and

Table 4. Distribution of EMS activations by complaint reported by dispatch.

Complaint Reported by Dispatch ¹	Overall N = 19,533,036
Sick Person	3,872,663 (20%)
Breathing Problem	2,207,589 (11%)
Falls	1,811,338 (9%)
Chest Pain (Non-Traumatic)	1,315,271 (7%)
No Other Appropriate Choice	1,157,885 (6%)
Unconscious/Fainting/Near-Fainting	953,687 (5%)
Traffic/Transportation Incident	902,210 (5%)
Abdominal Pain/Problems	762,783 (4%)
Psychiatric Problem/Abnormal Behavior/Suicide Attempt	725,938 (4%)
Convulsions/Seizure	652,807 (3%)
Unknown Problem/Person Down	608,769 (3%)
Traumatic Injury	557,312 (3%)
Transfer/Interfacility/Palliative Care	531,974 (3%)
Stroke/CVA	496,445 (3%)
Overdose/Poisoning/Ingestion	461,083 (2%)
Hemorrhage/Laceration	411,887 (2%)
Diabetic Problem	259,613 (1%)
Back Pain (Non-Traumatic)	252,937 (1%)
Heart Problems/AICD	246,602 (1%)
Pandemic/Epidemic/Outbreak	202,204 (1%)
Cardiac Arrest/Death	200,810 (1%)
Assault	197,900 (1%)
Pregnancy/Childbirth/Miscarriage	131,333 (1%)
Allergic Reaction/Stings	130,764 (1%)
Headache	99,723 (1%)
Stab/Gunshot Wound/Penetrating Trauma	95,596 (<1%)
Medical Alarm	61,767 (<1%)
Choking	36,593 (<1%)
Well Person Check	31,866 (<1%)
Heat/Cold Exposure	26,140 (<1%)
Animal Bite	24,862 (<1%)
Burns/Explosion	23,242 (<1%)
Airmedical Transport	20,028 (<1%)
Eye Problem/Injury	15,743 (<1%)
Fire	8,771 (<1%)
Standby	7,152 (<1%)
Automated Crash Notification	6,459 (<1%)
Carbon Monoxide/Hazmat/Inhalation/CBRN	6,192 (<1%)
Industrial Accident/Inaccessible Incident/Other Entrapments (Non-Vehicle)	5,996 (<1%)
Drowning/Diving/SCUBA Accident	5,012 (<1%)
Health care Professional/Admission	3,485 (<1%)
Electrocution/Lightning	2,605 (<1%)

¹eDispatch.01-Complaint Reported by Dispatch.

Results reflect data as of 2020 public data, treat and transported 9-1-1 response.

patient care information by types of agencies, patient populations and injury/illness types (17).

Currently, there are 47 states among the 50 states and territories submitting data to the National EMS Database that indicate greater than 90% of all 9-1-1 EMS ground activations are captured in state datasets. Nearly all the remaining states report submitting greater than 75% of 9-1-1 EMS ground activations. It is of interest to note that no state or territory, once submitting version 3 data to the National EMS Database, has failed to submit in a following year.

The National EMS Database represents a registry of EMS activations, rather than a registry of patients receiving care. This particularity is due to the fact that multiple emergency resources responding to the same 9-1-1 call may submit a PCR to the state data repository. Methods are not currently in place, at the state or national level, to link different PCRs to the same patient or emergency event. Accordingly, the National EMS Database should be considered a registry of EMS activations or responses, rather than patients. The database does include elements that characterize the role of the EMS unit for each specific call (e.g., eResponse.07-

Table 5. Distribution of EMS Activations by Primary Symptom (Top 20 results).

Primary Symptom ¹	Overall N = 19,533,036
Asthenia	1,377,482 (7%)
Shortness of breath	1,293,095 (7%)
Generalized pain	1,220,962 (6%)
Chest pain, unspecified	919,516 (5%)
Change in mental status	898,504 (5%)
Other general symptoms and signs	572,126 (3%)
Back pain	460,139 (2%)
Generalized abdominal pain	348,536 (2%)
Hemorrhage	342,861 (2%)
Headache	313,830 (2%)
Light-headedness	292,150 (1%)
Syncope and collapse	287,006 (1%)
Nausea	275,254 (1%)
Fever with chills	265,483 (1%)
Chronic debility	263,826 (1%)
Strange and inexplicable behavior	224,716 (1%)
Seizure (unspecified)	195,669 (1%)
Encounter for observation	193,343 (1%)
Vomiting, unspecified	191,569 (1%)
All other primary symptoms combined	7,869,279 (40%)
Missing	1,727,690 (9%)

¹eSituation.09-Primary Symptom.

Results reflect data as of 2020 public data, treat and transported 9-1-1 response.

Table 6. Distribution of EMS activations by primary impression (top 20 results).

Primary Impression ¹	Overall N = 19,533,036
Asthenia	1,174,186 (6%)
Generalized abdominal pain	986,134 (5%)
Chest pain, unspecified	962,269 (5%)
Injury, unspecified	925,256 (5%)
Shortness of breath	807,215 (4%)
Change in mental status	806,108 (4%)
Acute pain, not elsewhere classified	640,830 (3%)
Mental illness	435,714 (2%)
Syncope and collapse	420,204 (2%)
Back pain	362,960 (2%)
Chronic debility	343,329 (2%)
Generalized pain	302,348 (2%)
Unspecified injury of head	254,941 (1%)
Cerebral infarction, unspecified	241,432 (1%)
Light-headedness	233,475 (1%)
Epileptic Seizures	233,021 (1%)
Symptoms and signs involving emotional state	207,660 (1%)
Seizure (unspecified)	206,799 (1%)
Cardiac arrhythmia, unspecified	202,897 (1%)
All Other Primary Impressions Combined	8,523,180 (44%)
Missing	1,263,078 (6%)

¹eSituation.11-EMS Clinicians Primary Impression.

Results reflect data as of 2020 public data, treat and transported 9-1-1 response.

Primary Role of the Unit). Therefore, if an EMS response system is understood for a region, NEMSIS data could be stratified to better characterize emergency events or patients. It is also plausible that, when considering data relationships (e.g., ratios), database findings may approximate those based solely on patient or activation counts. For example, if considering the percentage of patients that are “female”, there is little reason to believe that the probability of a specific EMS response structure (e.g., single-tier versus dual-tier) would systematically differ by patient sex. Thus, if the bias is unrelated, ratios may be less affected.

Another common concern regarding the National EMS Database is the idea that states have differing inclusion criteria for submitting EMS activations. That is, states including EMS activations that result in a canceled call or “no patient found” will introduce a sampling bias when compared to

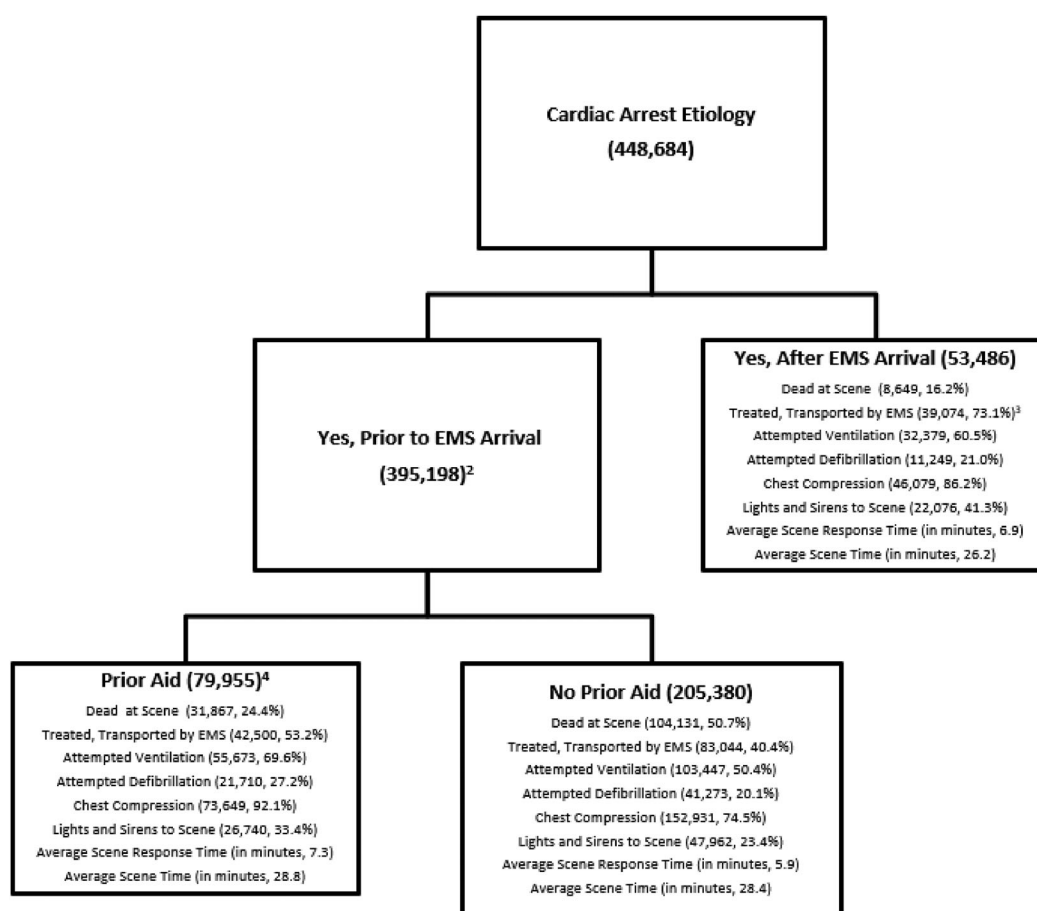


Figure 3. Out-of-Hospital Cardiac Arrest EMS Report¹.

¹Included all EMS activations in which an out-of-hospital cardiac arrest was observed regardless of transport.

²109,863 activations with unknown status of aid.

³Other dispositions included refusals, assists, and other transport methods (5,763, 10.7%).

⁴Prior aid included interventions performed by law enforcement, lay person, or other clinician.

Table 7. Distribution of EMS Activations by cause of injury for injured patients (Top 20 Results).

Cause of Injury ¹	Overall N = 3,568,704
Fall on same level, slip, or trip	1,328,616 (37%)
Motor Vehicle Collision	616,242 (17%)
Assault	246,234 (7%)
Fall from object	217,963 (6%)
Fall from one level to another	145,935 (4%)
Unspecified events with undetermined intent	61,033 (2%)
Contact with sharp object, undetermined intent	45,814 (1%)
Intentional self-harm	40,929 (1%)
Motorcycle accident	39,173 (1%)
Pedestrian injured in collision with car, pick-up truck or van in traffic accident	35,849 (1%)
Striking against or struck by other objects	30,792 (1%)
Pedal cyclist injured in traffic accident	24,284 (1%)
Accidental hit, strike, kick, twist, bite or scratch by another person	13,317 (<1%)
Caught, crushed, jammed or pinched in or between objects	9,935 (<1%)
Struck by thrown, projected or falling object	8,421 (<1%)
Firearm discharge, undetermined intent	8,025 (<1%)
Vehicle collision with fixed or stationary object	7,189 (<1%)
Contact with dog	6,944 (<1%)
Contact with machinery	6,842 (<1%)
All other causes of injury combined	342,894 (10%)
Missing	332,273 (9%)

¹eInjury.01-Cause of Injury and eSituation.02-Possible Injury is "Yes". More than one cause of injury may be recorded per PCR. Causes of Injury with the same ICD-10 parent code were combined.

Results reflect data as of 2020 public data, treat and transported 9-1-1 response.

states requiring patient contact for inclusion. The extent of this potential bias can be assessed since the database includes elements characterizing the disposition of the initial EMS response. All submitting states, except one, include all EMS activations resulting in any form of patient contact.

It is important to note that missing data are present and varied across different data elements in the NEMSIS dataset. In most cases, data elements are not missing at random and analyses, therefore, are subject to bias if missing data are ignored. Under circumstances in which missing data might bias an analysis, several different approaches could be taken to minimize the effect. Single or multiple imputation techniques attempt to "mimic" missing data by providing an "informed guess" at a value, substitution with a mean value, or substitution based on a regression equation or multiple simulations using other (observed) values (18). Another straightforward approach to dealing with missing data is to rely on covarying elements to inform the analysis. For example, one might assume that a patient has suffered an injury, even if the element identifying a possible injury (eSituation.02-Possible Injury) reports a null value; the clinician can indicate a value in eInjury.01-Cause of Injury and document some form of trauma in eSituation.11-EMS Clinicians Primary Impression. Regardless of the approach taken, it is important to explore the impact of attempts to

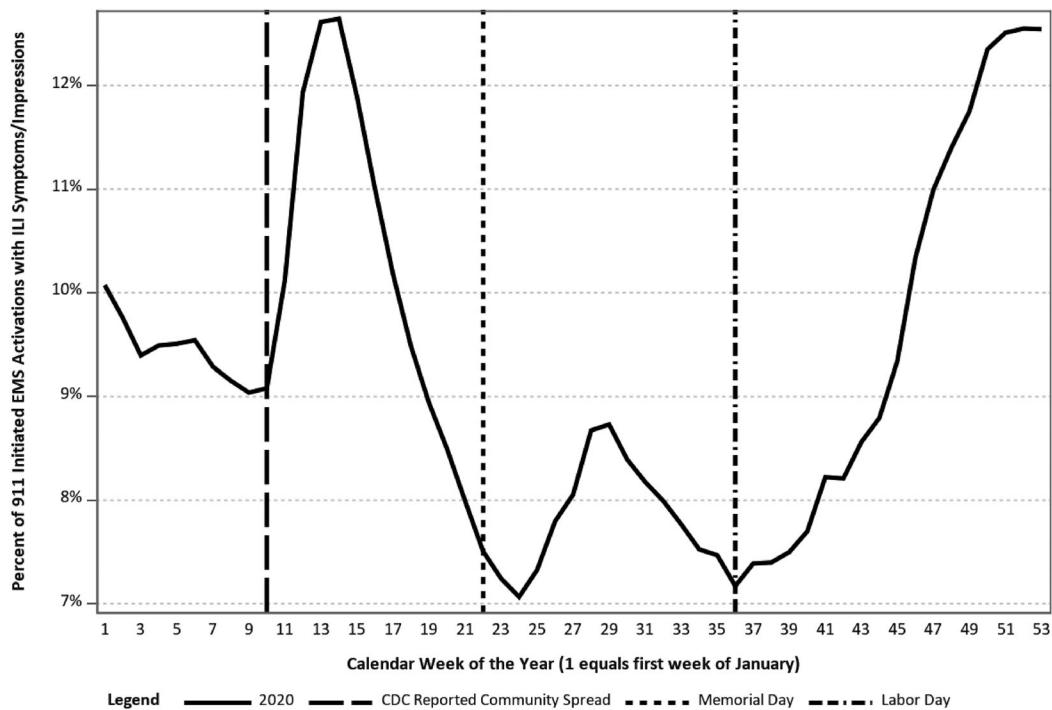


Figure 4. Percent of 9-1-1 Initiated EMS Activations with ILI Symptoms/Impressions by Week*.

*Percent calculated as: Numerator – patients presenting with Influenza-Like Illness (ILI) symptoms documented as a patient symptom or EMS clinician impression. Denominator – 9-1-1 initiated EMS activations with patient contact.

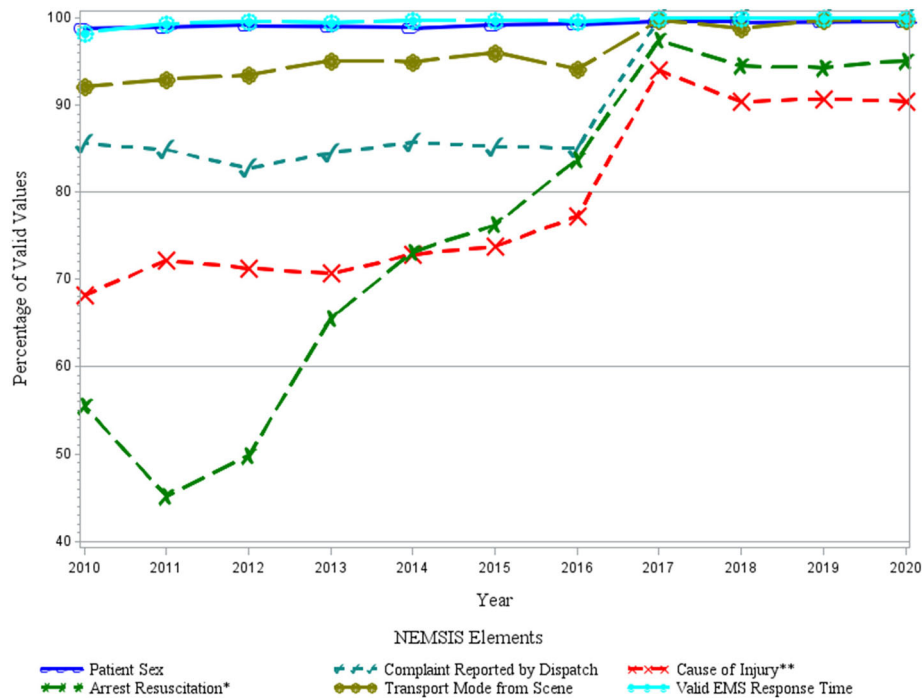


Figure 5. Validity of PCR values over time.

*Includes only cases where cardiac arrest occurs after EMS arrival.

**Includes only cases where a possible injury is indicated.

resolve missing data using accepted methods, such as sensitivity analyses (19).

Conclusion

The National EMS Database encompasses the largest publicly available prehospital clinical health care dataset in the

U.S. By understanding the advantages and limitations associated with using the NEMSIS dataset, meaningful contributions to the improvement of prehospital emergency response, patient care and transport can be made. As the National EMS Database includes EMS data from 2007 to present day, the related findings from the NEMSIS dataset can be analyzed for trends in response variations over time

to increase the significance of temporal changes and impacts to the EMS system. We encourage the full utilization of this dataset as well as EMS data housed at local and state levels that contribute to this national project.

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